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Knowledge Check: Language Models
Practice Assignment • 10 min

Due Oct 26, 11:59 PM IST

Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item

1. Why do LLMs with more parameters tend to perform better? 1 / 1 point

- (•) They capture richer representations of language and context
- ( ) They store static predefined outputs
- ( ) They require fewer training iterations
- ( ) They use less data for training

Nice work

Correct! More parameters enable richer modeling of language.

2. What is a drawback of using *unigram models*? 1 / 1 point

- (•) They produce text that lacks meaningful structure or context
- ( ) They require advanced GPU hardware
- ( ) They cannot predict any words at all
- ( ) They always generate grammatically correct sentences

Nice work

Correct! Unigram models do not account for context, leading to less coherent output.

3. Which of the following can be used to build a language model? 1 / 1 point

- [ ] Only single sentence length inputs
- [x] Neural network architectures

Nice work

Correct! Modern language models are frequently based on neural networks.

- [ ] Manual grammar rules only
- [x] Counting word sequences in a large corpus

Nice work

Correct! Language models are often built by analyzing word sequence statistics.

4. What happens when you use a *trigram model* to generate text? 1 / 1 point

- ( ) It only generates question marks
- ( ) It always generates numerical data
- ( ) It uses images instead of text
- (•) It predicts the next word based on the previous two words

Nice work

Correct! A trigram model uses the last two words as context.

5. What is the primary purpose of a language model? 1 / 1 point

- ( ) To remove stop words from text
- (•) To predict the next word given a sequence of previous words
- ( ) To sort words alphabetically
- ( ) To translate text between languages

Nice work

Correct! A language model predicts the next word based on prior context.

6. What are emergent abilities in LLMs? 1 / 1 point

- ( ) Features that require manual programming
- (•) Capabilities that appear in large models but not in smaller models
- ( ) Abilities that only appear in rule-based systems
- ( ) Features that degrade model performance

Nice work

Correct! Emergent abilities show up when models grow sufficiently large.

7. What kind of training loss is used to train LLMs? 1 / 1 point

- ( ) Spectral loss
- ( ) Hinge loss
- ( ) Mean squared error loss
- (•) Cross-entropy loss

Nice work

Correct! Cross-entropy loss is the standard loss used to train LLMs.